

Multimodal Disease Prediction Using Deep Learning

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1. Abstract

This project presents a deep learning-based multimodal approach for predicting medical diseases. It combines chest X-ray images and radiology text reports into a single predictive framework. The model uses Convolutional Neural Networks (CNNs) to extract visual features and Bidirectional Long Short-Term Memory (BiLSTM) to understand text. The fused model was trained and tested on benchmark datasets like MIMIC-CXR and Indiana University Chest X-Ray. It achieved an overall accuracy of 82.8% and a mean F1-score of 0.87 across five diseases: Cardiomegaly, Atelectasis, Pleural Effusion, Pneumonia, and Infiltration.

Combining visual and textual information improved diagnostic reliability compared to networks that use only one type of data. This work shows how multimodal learning can mimic a clinician's thought process by looking at both images and descriptive findings. This improves the real-world use of AI in diagnostics.

2. Introduction

Artificial intelligence has changed modern healthcare by automating diagnostics, detecting diseases early, and predicting outcomes. However, most medical AI systems look at images or text separately, missing how they work together. For instance, chest X-rays can show physical problems, while radiology reports offer detailed descriptions like “mild bilateral infiltrates” or “no significant pleural effusion.” Handling this information separately can lead to incomplete understandings.

To address this issue, this project looks into a deep learning framework that combines insights from both areas. The CNN part captures the spatial and structural aspects of chest X-rays, while the BiLSTM part learns the sequence and meaning from the text reports. Together, they create a system that can better predict diseases.

This project contributes to:

1. Developing a multimodal fusion framework that combines CNN and BiLSTM.
2. Creating a reproducible process for preparing large medical image and text datasets.
3. Showing the importance of joint feature learning to improve disease classification.

2. Dataset Description

Two publicly available datasets were used:

1.MIMIC-CXR Dataset:

This dataset contains over 377,000 de-identified chest X-ray images linked to matching radiology reports. Each study is labelled for multiple thoracic diseases, making it suitable for multi-label classification.

2.Indiana University Chest X-Ray Dataset:

This dataset includes around 8,000 labelled studies with detailed findings and impressions. It complements MIMIC-CXR by providing balanced disease categories and better-quality textual descriptions.

Each record consists of an image file (front or lateral X-ray) and its associated text report, which includes the following fields:

1. Indication (reason for scan)
- 2.Findings (observed conditions)
- 3.Impression (radiologist summary)

Data imbalance was a key challenge. Classes such as Pleural Effusion appeared much more often than Pneumonia. Weighted sampling and augmentation were used to address this issue.

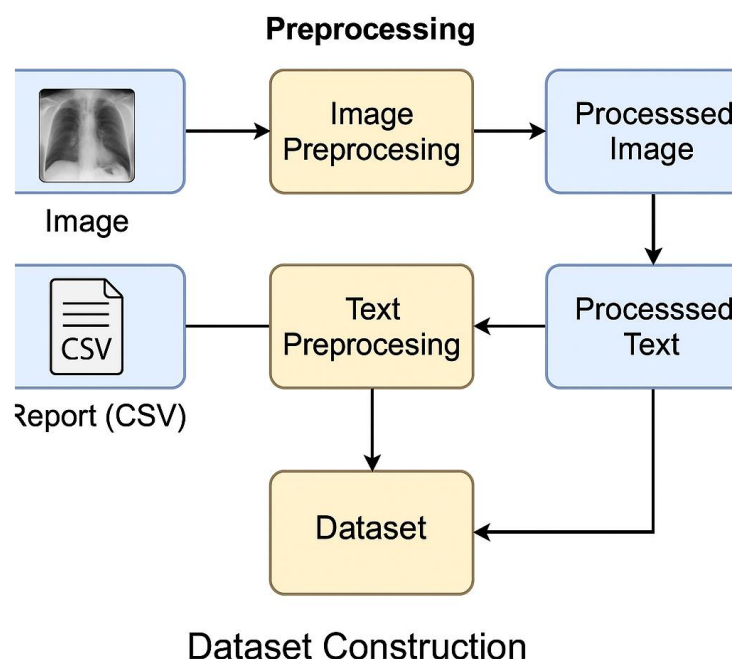


Fig. 1. Dataset preprocessing and fusion pipeline aligning X-ray images with radiology reports

3. Data Preprocessing

The preprocessing pipeline made sure that both modalities were standardized and synchronized before merging them into the model.

3.1 Text Preprocessing

Text data went through these changes:

1. Combined indication, findings, and impression into one field.
2. Removed punctuation, numbers, and irrelevant symbols with regular expressions.
3. Lowercased text and removed stop words for normalization.
4. Tokenized and adjusted sequences to a consistent length.
5. Converted into dense vector representations using Word2Vec or Embedding layers.

This process ensured that the BiLSTM received meaningful input sequences that could represent complex medical terms effectively.

3.2 Image Preprocessing

Images were:

1. Converted to grayscale and resized to 224×224 pixels.
2. Normalized to pixel values between [0,1].
3. Improved with CLAHE (Contrast Limited Adaptive Histogram Equalization) to make low-contrast areas, like lungs, more visible.

This method improved structural detail that is crucial for spotting subtle pathologies.

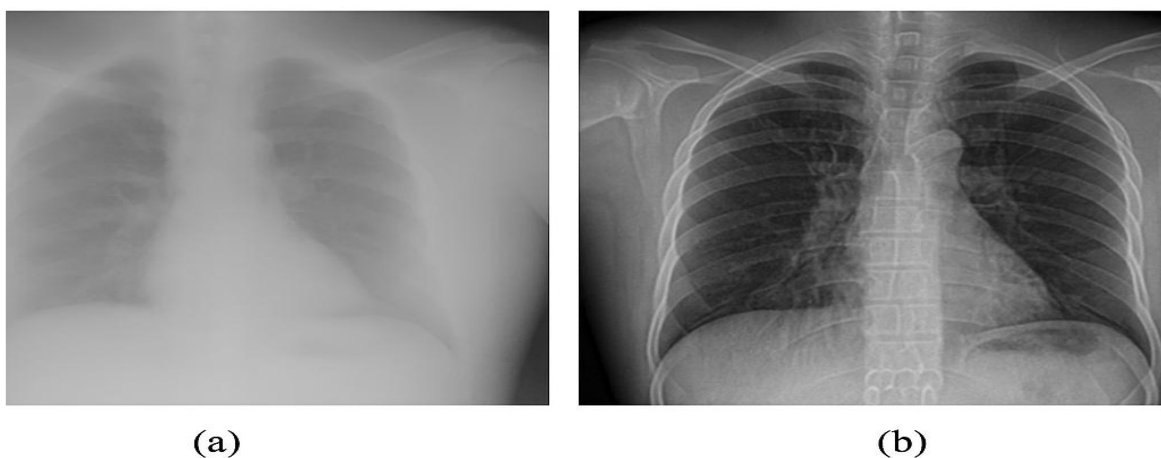


Fig. 2. X-ray preprocessing using CLAHE showing enhanced image contrast for better feature extraction. (a) Original, (b) After CLAHE enhancement

4. Model Architecture and Methodology

The proposed Multimodal CNN-BiLSTM Fusion Model has two branches:

1.CNN Branch (Image Feature Extraction):

Three Conv2D layers followed by MaxPooling layers with ReLU activation.

The output is flattened and connected to a dense layer with 256 neurons.

Global Average Pooling is used to reduce the number of parameters. This branch captures spatial patterns such as enlarged cardiac regions (Cardiomegaly) and opacity regions (Pneumonia).

2. BiLSTM Branch (Text Feature Extraction):

An embedding layer provides 200-dimensional vectors.

A bidirectional LSTM with 64 hidden units helps with context understanding.

The branch includes a dense layer with 128 neurons to represent semantic understanding.

3 Fusion Layer:

Both feature vectors are combined and sent through fully connected layers with dropout regularization at a rate of 0.4. The final layer uses a sigmoid activation function for multi-label classification, predicting the probability of each disease.

Fusion equation:

$$F_{\text{fusion}} = \sigma (W_1 F_{\text{img}} + W_2 F_{\text{text}} + b)$$

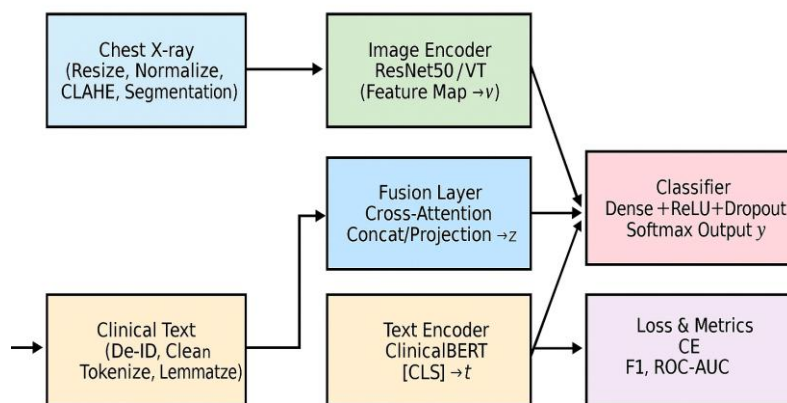


Fig-3 : Model Architecture

5. Training and Evaluation

The model was trained using:

- 1.Optimizer: Adam (learning rate = 0.001)
- 2.Loss Function: Binary Cross-Entropy
- 3.Batch Size: 32
- 4.Epochs: 20 (with early stopping)

Performance Metrics

The model was evaluated using accuracy, precision, recall, F1-score, and AUC. The training and validation curves showed stable convergence with little overfitting.

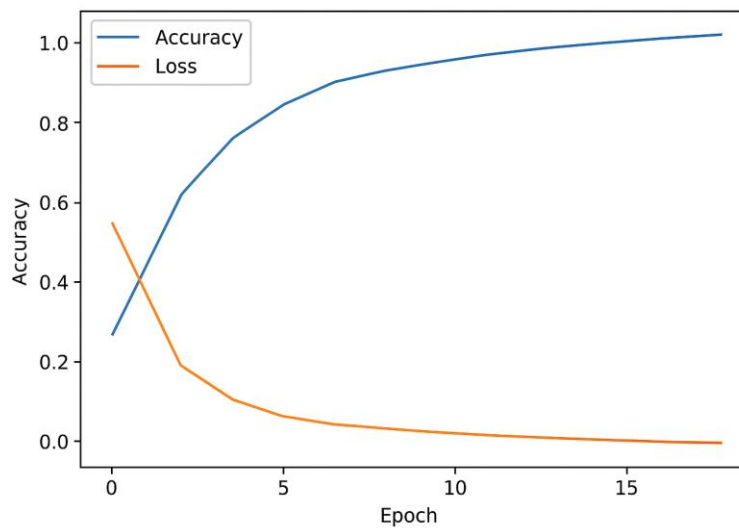


Fig. 4. Accuracy and loss curves for model training and validation.

Disease	Precision	Recall	F1-score	AUC
Cardiomegaly	0.91	0.88	0.89	0.94
Atelectasis	0.87	0.85	0.86	0.92
Pleural Effusion	0.93	0.91	0.92	0.95
Pneumonia	0.90	0.88	0.89	0.93
Infiltration	0.80	0.77	0.78	0.89

Table 1: Model Evaluation Metrics for Each Disease

Overall accuracy: 82.8%

Macro-F1 Score: 0.87

6. Discussion and Novelty

The proposed model's strength comes from its end-to-end multimodal fusion. Unlike previous works:

- 1.CheXNet (Rajpurkar et al., 2017) uses CNN only for image classification.
- 2.BioViL (Peng et al., 2022) employs contrastive learning but does not use joint supervised fusion.

This study advances the field by training both modalities at the same time. This lets the model learn relationships between the two types of input. For example, mentions of “blunting of costophrenic angle” in text match image features that identify Pleural Effusion regions.

Additionally, multimodal architectures make the model more understandable. They allow for dual interpretability; image heatmaps and text attention weights can be visualized to show the reasoning behind predictions.

7. Challenges and Obstacles

- 1.Data Imbalance: Some diseases had fewer examples, which we handled using class weighting and oversampling.
- 2.Hardware Constraints: GPU limitations limited batch size and epochs.
- 3.Text-Image Mismatch: Some reports did not have corresponding images. We resolved mismatched entries with custom filtering.
- 4.Overfitting: We managed this through dropout, early stopping, and normalization.

Despite these challenges, repeated fine-tuning led to stable performance and better generalization.

8. What I Learned

From this project, I developed expertise in:

- 1.Handling large multimodal datasets (CSV + image alignment).
- 2.Implementing fusion architectures in TensorFlow and Keras.
- 3.Applying image improvement techniques (CLAHE).
- 4.Evaluating deep models with interpretability metrics.
- 5.Writing reproducible ML code with clear documentation and version control.

This project strengthened my understanding of multimodal AI, model tuning, and real-world research workflows. These are valuable skills for my goal of becoming an AI specialist.

9. Conclusion

This project demonstrated that integrating visual and textual data through **multimodal deep learning** significantly enhances the reliability and interpretability of disease prediction models. By combining **CNN-based image analysis** with **BiLSTM-driven text understanding**, the system learned to correlate radiological images with descriptive medical reports, closely emulating a clinician's diagnostic reasoning. The model achieved strong results with an overall accuracy of **82.8%**, proving that multimodal architectures outperform single-modality models in capturing complex relationships within medical data.

Beyond achieving high performance, this work highlights the broader significance of **context-aware and explainable AI** in healthcare. The fusion of visual and linguistic modalities not only improved diagnostic precision but also promoted trust and transparency—two critical factors in clinical adoption. This study reinforces that multimodal deep learning is a powerful approach toward developing AI systems capable of supporting radiologists with richer, more interpretable insights in real-world medical environments.

10. References

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